

LCA of impossible - Myths & Legends

MJÖLNIR

WHEN THUNDER BECOMES THE PRODUCT.

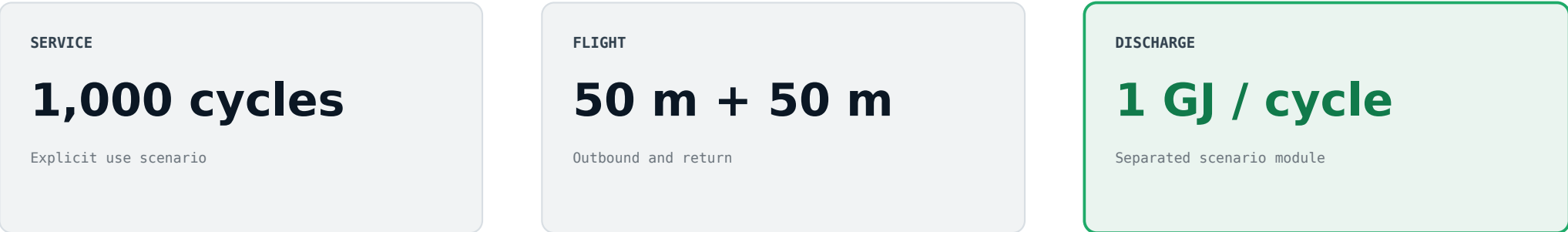


STEP ONE

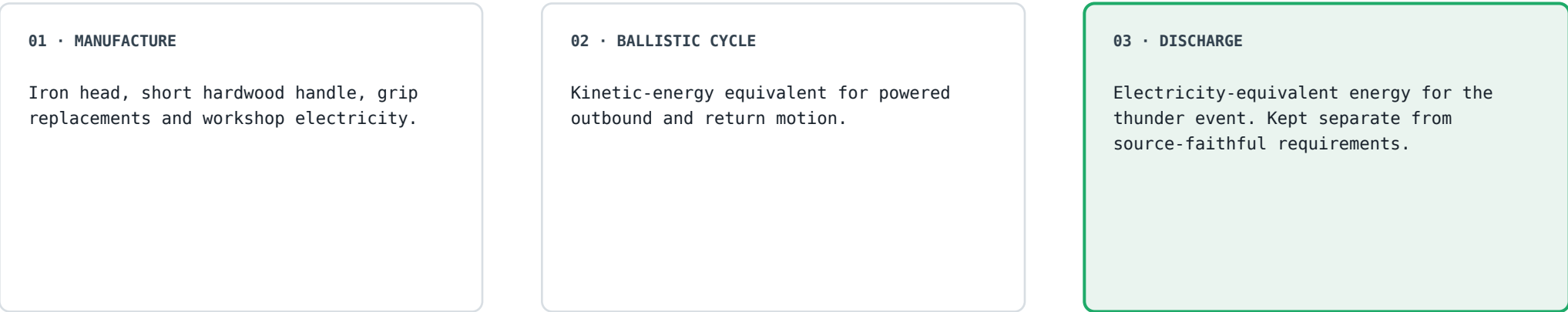
Function before footprint

FUNCTIONAL UNIT

Manufacture and deployment of one reusable Mjöltnir-class iron hammer over 1,000 combat cycles. Each cycle contains a 50 m outbound flight, a 50 m powered return and one 1 GJ lightning-equivalent discharge.



THREE CALCULATION MODULES



The denominator is intentionally operational. A one-off product footprint would describe the hammer; the combat-cycle basis tests what makes it Mjöltnir.

STEP ONE

What the source requires

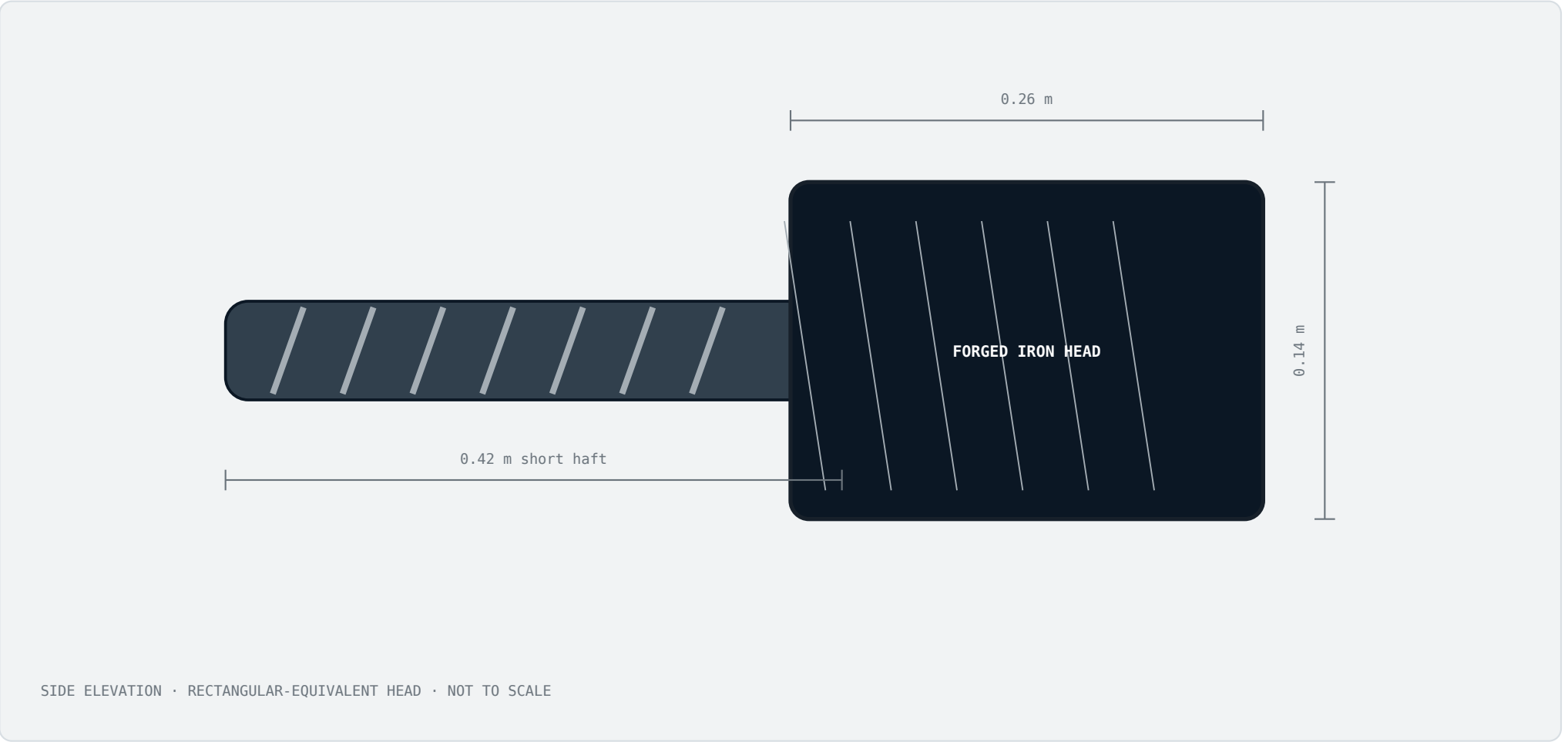
SOURCE REQUIREMENT	MODEL RESPONSE	STATUS
Sindri forges the head from iron	Rectangular-equivalent iron head; primary-metal screening proxy	Modeled
A blow as hard as desired will not make it fail	Durable reusable head; no damage replacement over 1,000 cycles	Modeled
When thrown, it does not miss	Accuracy/control energy not quantified	Excluded
It returns to the hand	Powered return represented by kinetic-energy equivalent	Modeled
The fore-haft is short	0.42 m short hardwood handle	Modeled
Electrical lightning discharge	Not stated in this passage; introduced only as a separate scenario	Scenario

EVIDENCE RULE

The ancient text defines performance. Dimensions, velocities, efficiencies and service cycles are modern engineering assumptions. Lightning is a deployment scenario, not smuggled in as a literary fact.

STEP TWO

A compact 46.7 kg machine



FINISHED HEAD

45.72 kg

0.005824 m3 iron

HANDLE + GRIP

1.00 kg

Initial configuration

FINISHED TOOL

46.72 kg

Before replacements

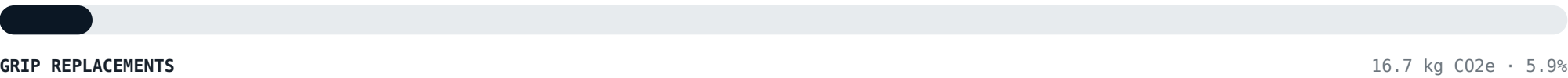
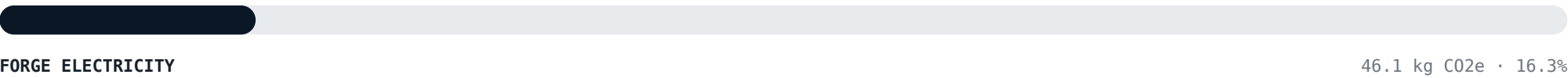
MANUFACTURING ASSUMPTIONS

- 80% forging yield converts the 45.72 kg finished head into 57.15 kg of metal input.
- 70% woodworking yield and two replacement sets produce 3.21 kg of wood input over the functional unit.
- Forge and finishing electricity: 250 kWh.

STEP TWO

Manufacturing: 0.282 t CO2e

FLOW	ACTIVITY	FACTOR	GWP
Primary metal	0.05715 t	3,821.95 kg CO2e/t	218.42 kg
Wood: 3 handles	0.00321 t	269.50 kg CO2e/t	0.86 kg
Grip: 3 sets	0.00075 t	22,310 kg CO2e/t	16.73 kg
Forge electricity	250 kWh	0.18436 kg CO2e/kWh	46.09 kg



MANUFACTURING MODULE

282 kg CO2e

Before any flight or thunder

STEP TWO

The return is measurable

KINETIC ENERGY PER LEG

$$E = \frac{1}{2}mv^2 = 0.5 \times 46.7168 \text{ kg} \times (50 \text{ m/s})^2 = 58.40 \text{ kJ}$$

POWERED OUTBOUND + RETURN

$$E_{\text{cycle}} = 2 \times 58.40 \text{ kJ} / 0.60 = 194.65 \text{ kJ} = 0.0541 \text{ kWh}$$

ACROSS THE FUNCTIONAL UNIT

$$0.0541 \text{ kWh/cycle} \times 1,000 \text{ cycles} = 54.07 \text{ kWh}$$

FLIGHT EMISSIONS

9.97 kg CO2e

0.00997 kg per cycle

SHARE OF TOTAL

0.019%

Kinetic energy is not the hotspot

Even doubling speed to 100 m/s only quadruples this module to 39.9 kg CO2e. The return looks supernatural, but under the stated energy-equivalent model it remains numerically small.

STEP TWO

Thunder sets the order of magnitude

ENERGY CONVERSION

1 GJ ÷ 3.6 MJ/kWh = 277.78 kWh per discharge

PER COMBAT CYCLE

277.78 kWh

1 GJ electricity equivalent

1,000 CYCLES

277.78 MWh

Before applying the factor

CARBON CALCULATION

277,777.78 kWh × 0.18436 kg CO2e/kWh = 51,211 kg CO2e

INTERPRETIVE SEPARATION

- This is an electricity-equivalent accounting device, not a claim that natural lightning is generated by the UK grid.
- The 1 GJ main case is the lower bound of the published 1–10 GJ order of magnitude.
- A 0.1 GJ low engineering case is retained to show model sensitivity below that published range.

STEP THREE

A modern proxy stack

FULL ELECTRICITY-CONSUMPTION PROXY

COMPONENT	KG CO2E/KWH	BOUNDARY
UK electricity generation	0.13096	Generation
Transmission & distribution	0.01299	Network losses
WTT – generation	0.03682	Upstream fuel / energy
WTT – T&D;	0.00359	Upstream network
COMBINED	0.18436	Consumption equivalent

EXCLUDED

INCLUDED

- Primary metal, wood and grip material production
 - Forge/workshop electricity
 - Powered flight equivalent
 - Lightning-equivalent discharge
- Specific mining and transport
 - Charcoal and direct forge fuels
 - Thor, targets, sound and heat
 - Recovery and end of life

REPRESENTATIVENESS

DEFRA 2026 is strong for factor transcription and weak for Asgard. The workbook keeps source quality separate from arithmetic precision.

STEP FOUR

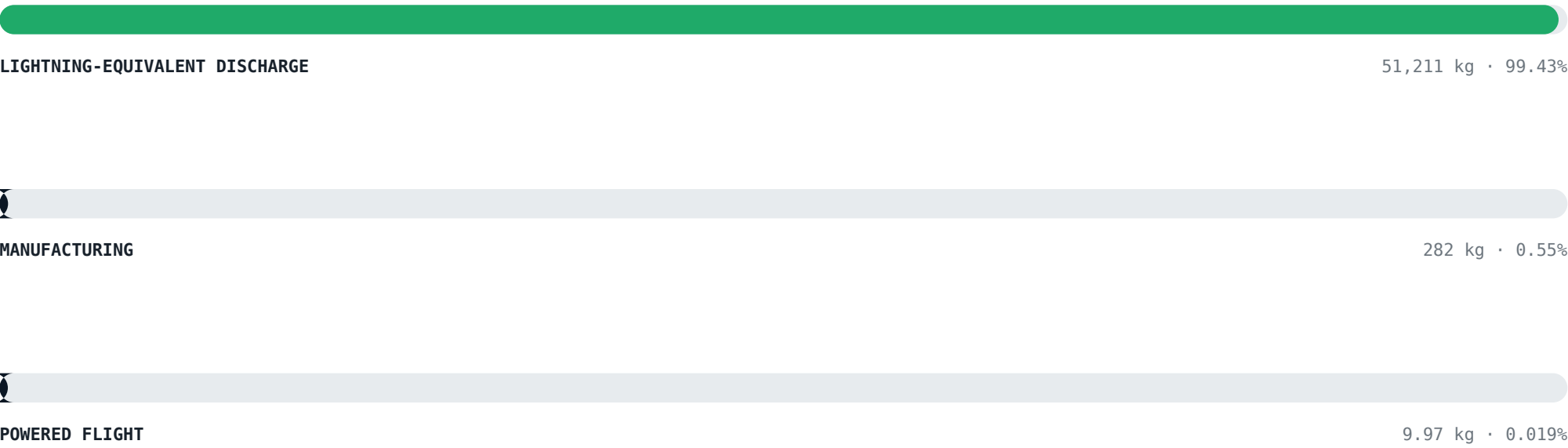
51.5 t CO2e per functional unit

MAIN RESULT

51.50 t CO2e

51.50 kg CO2e per combat cycle

CONTRIBUTION BY MODULE



HOTSPOT STATEMENT

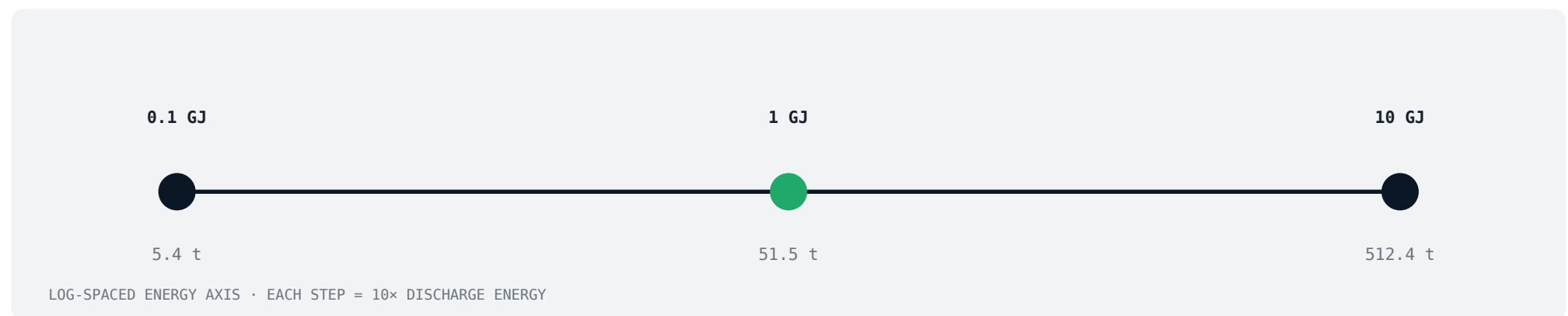
The head is less than 1% of the product-system footprint. Thunder is the product.

The physical hammer still matters for engineering credibility and durability. It does not govern the modeled climate result once a gigajoule-scale event is repeated 1,000 times.

STEP FIVE

One parameter moves two orders

CASE	GJ / CYCLE	TOTAL	VS MAIN
No discharge	0	0.292 t CO2e	-99.43%
Low engineering	0.1	5.413 t CO2e	-89.49%
Main	1	51.503 t CO2e	baseline
Upper lightning	10	512.403 t CO2e	+894.89%



DECISION CONSEQUENCE

Before debating iron sourcing, define what one discharge means. A one-order change in event energy creates an almost one-order change in total GWP.

STEP FIVE

The residual is robust

SCENARIO	MANUFACTURE	FLIGHT	TOTAL	CHANGE
Primary metal · 50 m/s	282.1 kg	10.0 kg	51.503 t	baseline
Closed-loop metal · 50 m/s	157.2 kg	10.0 kg	51.378 t	−0.24%
Primary metal · 25 m/s	282.1 kg	2.5 kg	51.496 t	−0.015%
Primary metal · 100 m/s	282.1 kg	39.9 kg	51.533 t	+0.058%

CLOSED-LOOP METAL

−125 kg CO2e

Manufacturing module only

TOTAL SYSTEM EFFECT

−0.24%

Discharge unchanged

TECHNICAL READING

- Metal circularity cuts the physical product footprint by 44%, from 282 kg to 157 kg CO2e.
- At system level, the same improvement is almost invisible because the discharge remains fixed.
- Speed changes flight energy quadratically, but flight is too small to alter the headline order of magnitude.

STEP SIX

Engineer the thunder first

FINDING 01

A 46.7 kg reusable hammer can be manufactured for roughly 0.282 t CO2e under the selected modern proxies.

FINDING 02

The powered flight-and-return equivalent adds only 9.97 kg CO2e over 1,000 cycles.

FINDING 03

At 1 GJ per cycle, the discharge adds 51.21 t CO2e and determines 99.43% of the result.

DESIGN PRIORITY

- Decarbonize, recover or redefine the discharge energy before optimizing material mass.
- Keep source-faithful requirements and culturally familiar additions in separate modules.
- Report the physical product footprint alongside the operational scenario so the conclusion remains interpretable.

Mjöltnir is an energy system with a hammer-shaped housing.

LCA of the impossible - Season II

MYTHS & LEGENDS

**Where ancient stories meet
measurable emissions**